

Erik Löffelholz

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PERSONAL PROFILE & RESEARCH INTERESTS

Master's graduate in Mathematical Physics with a strong foundation in continuous–discrete interactions and differential geometry. Currently specializing in Graph Machine Learning and physics-inspired generative modeling. Passionate about understanding how complex structure emerges from local constraints by integrating variational principles, energy-based modeling, and structural priors. Applied experience spans joint discrete–continuous diffusion models, differentiable physics simulation, discrete differential geometry, mesh embeddings into curved spaces, rigorous metric evaluation, and from-scratch gradient-based optimization.

EDUCATION

Oct 2021 – Sep 2025	M.Sc. Mathematical Physics Universität Leipzig — Faculty of Physics and Earth System Sciences Selected Coursework: Advanced PDE and Analysis · Differential Geometry · Quantum Field Theory · General Relativity · Group Theory
Oct 2018 – Mar 2022	B.Sc. Physics Universität Leipzig — Faculty of Physics and Earth System Sciences
Aug 2010 – Jun 2018	Abitur Luther-Melanchthon-Gymnasium, Lutherstadt Wittenberg

RESEARCH EXPERIENCE

Apr 2025 – May 2026

Scientist

Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany

Joined as a student research assistant (Wissenschaftliche Hilfskraft) and promoted to Scientist for the final period (Apr–May 2026).

- Built GPU-accelerated tooling (PyTorch · Three.js) for embedding discrete meshes into Riemannian manifolds — Euclidean, spherical and hyperbolic spaces — via spring-mass energy minimization with analytic energy differentials; the computational basis for the co-authored Bridges 2026 paper on illustrating hyperbolic surfaces.
- Implemented a discrete differential geometry library: cotangent Laplacians, mass matrices, Gaussian/mean curvature, heat-method geodesic distances, isotropic remeshing and dihedral-angle bending energy across CUDA, MPS and CPU backends.
- Developed Ricci-flow surface evolution and procedural strand/weave generation on curved surfaces using half-edge topology, with real-time interactive 3D visualization.
- Engineered differentiable mesh-based physics simulators with a binary WebSocket protocol (10–20× faster than JSON) streaming PyTorch state to browser frontends.

2025 – Present

Independent Researcher & Developer

Graph ML, Computational Geometry & Differentiable Physics

- Built a from-scratch graph ML framework in pure PyTorch (no PyG/DGL) for spatial generation of 3D tree morphologies — joint discrete–continuous diffusion and autoregressive spatial-tree VAEs over graph topology and 3D node positions, with Sholl-analysis and spatial-MMD evaluation.
- Developed DiffQFT, a differentiable holographic QFT framework in Euclidean AdS_2 : Witten-diagram Monte Carlo integration through PyTorch autograd, neural surrogates and a PINN for the Klein–Gordon equation.
- Implemented a PINN solver for ten classical PDEs and browser-based differentiable-physics demos — a from-scratch neural ODE for chaotic Lorenz dynamics, and real-time particle/cloth simulation with an optional PyTorch backend.
- Built practical expertise with agentic coding models and LLM-assisted development workflows.

2025 – Present

Subject-Matter Expert — STEM & Coding

Outlier.ai — AI Training & Model Evaluation

- Author and review graduate-level mathematics, physics and science prompts and solutions for large-language-model training.
- Evaluate, rank and compare model outputs (RLHF) for correctness, reasoning quality and helpfulness.
- Complete coding and code-review tasks across Python, C++, JavaScript and front-end coding.
- Contribute German-language (de-DE) prompt and audio-prompt tasks.

TEACHING & ACADEMIC SERVICE

Mar 2024 – Feb 2025

Working Student — Mathematics Editorial

Ernst Klett Verlag GmbH, Leipzig

- Conducted technical and editorial reviews of mathematical content, ensuring structural clarity and pedagogical accuracy for educational publications.

PUBLICATIONS & CONFERENCES

Fabian Lander, **Erik Löffelholz**, Diaaeldin Taha, Steve Trettel, Anna Wienhard.

"Illustrating Hyperbolic Surfaces with Mesh Embeddings."

Submitted to the **Bridges Conference 2026** (Regular Papers Track). *Under Review*.

RELEVANT SKILLS & LANGUAGES

Programming

Python · JavaScript · TypeScript · C++

Scientific Computing

Numerical Integration (Runge-Kutta, Monte Carlo) ·
Differentiable Simulation · NumPy · SciPy

Tools & Workflow

Git · Docker · FastAPI · LaTeX · Agentic Coding Models ·
AI Data Labeling · RLHF / LLM Evaluation

Machine Learning

PyTorch (Autograd, PINNs, GNNs) · Diffusion Models ·
VAEs · Neural ODEs

Mathematical Methods

Functional analysis · PDE theory · variational methods ·
differential geometry · discrete differential geometry ·
group theory

Languages

German (native) · English (C1 — full professional
proficiency)

REFEREES

Dr. Diaaeldin Taha

Research Group Leader (Mathematical Structures in AI)

Max Planck Institute for Mathematics in the Sciences, Leipzig

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SELECTED PROJECTS

Mesh Embeddings & Discrete Differential Geometry — GPU mesh embedding into hyperbolic and spherical spaces via spring–mass energy minimization, with a discrete differential geometry operator library (cotangent Laplacian, curvature, heat-method geodesics). Computational basis for the Bridges 2026 paper.

Graph ML Lab — From-scratch implementations of GCN, GAT, graph VAE, discrete diffusion, and joint discrete-continuous diffusion over graph structure and 3D positions. PyTorch only, no external GNN libraries.

github.com/erik2810/ml-projects

DiffQFT — Differentiable quantum field theory computations in Euclidean AdS_2 . Witten diagram integration, neural surrogates, PINN solver for the Klein-Gordon equation. github.com/erik2810/DiffQFT

Differentiable Physics Engine — Neural ODE learning chaotic Lorenz dynamics from scratch in the browser. From-scratch backpropagation and Adam optimizer in JavaScript. github.com/erik2810/differentiable-physics-engine

Full project portfolio with live demos at erik2810.github.io/projects